WEEK 10 PROJECT RECORD

House Sales Analysis Dashboard using Power BI

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1. Introduction

This record is about my Week 10 project, where I made a Power BI dashboard on house sales data. The data has details of houses sold in and around Seattle, Washington, like the price, number of bedrooms and bathrooms, living area in square feet, grade, condition, waterfront and the location (latitude, longitude and zipcode) of each house.

The main aim of the project was to understand how the house price changes with different features of the house and where the houses are located. For this I made two report pages. The first page gives the overall numbers and a map with a living area slider. The third page shows the average house price compared with bedrooms, grade, bathrooms, condition and waterfront, and it has a price range slider to filter the data.

Page 1 has 7 visuals and the other page has 6 visuals. Each of them is explained below.

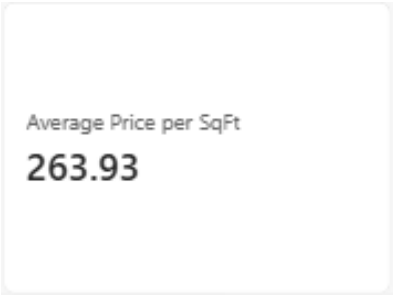
2. Dashboard Page 1 – Overview and Location

2.1 Average House Price (Card)



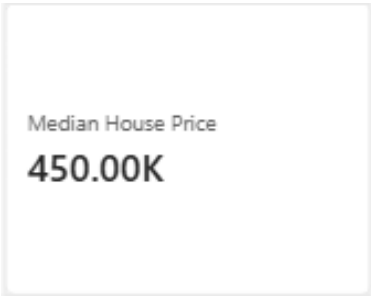
This card shows the average price of all the houses in the data, which is 540.65K. It gives a quick idea of what a normal house costs in this area.

2.2 Average Price per SqFt (Card)



This card shows the average price paid for one square foot of living space, which is 263.93. It helps to compare houses of different sizes on the same basis.

2.3 Median House Price (Card)



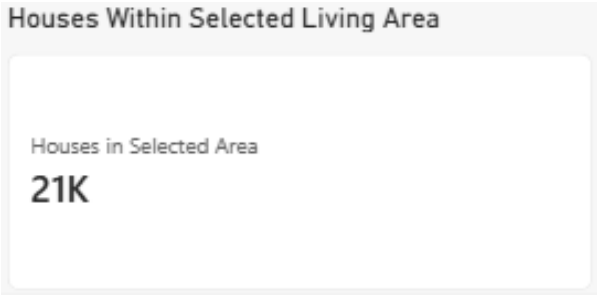
The median house price is 450.00K. It is lower than the average price, which means a few very costly houses are pulling the average upward.

2.4 Average Living Area (Card)



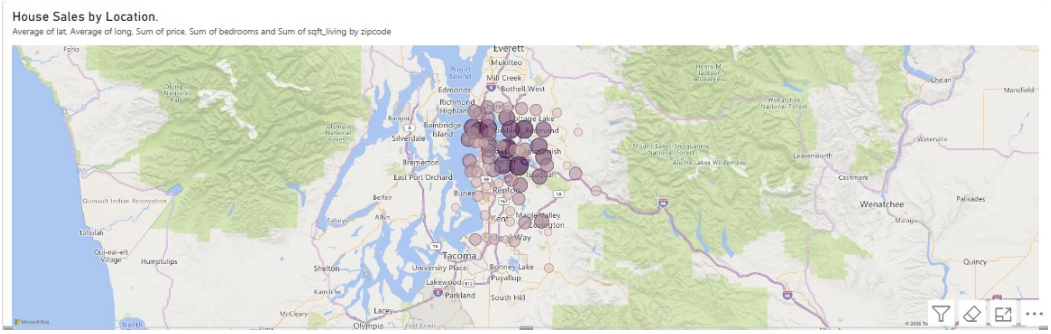
This card shows the average living area of a house, which is 2.08K sqft (about 2,080 square feet).

2.5 Houses in Selected Area (Card)



This card is under the heading "Houses Within Selected Living Area". It counts how many houses fall inside the living area chosen on the slider, and it shows 21K houses now. The number changes when the slider is moved.

2.6 House Sales by Location (Map)



This is a Bing map with bubbles placed using the latitude and longitude of the houses, grouped by zipcode. The size of a bubble depends on the total price, so bigger and darker bubbles mean higher sales. Most of the bubbles are around Seattle, Bellevue and Redmond, so we can see the sales are mostly near the Puget Sound area and very few are in the far east or south.

2.7 Living Area Range (Slider)



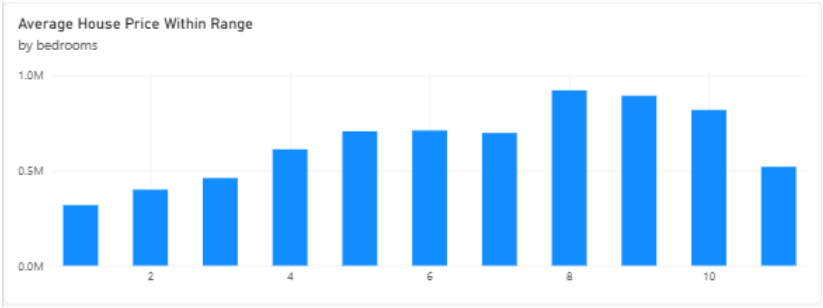
This slider is used to select the maximum living area in sqft (it is at 7000 in the picture). When we move it, the cards and the map update to show only the houses within that living area.

3. Dashboard Page 3 – Price Analysis



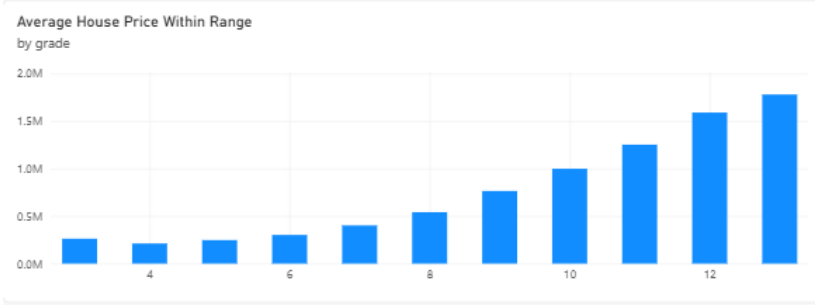
Figure 1: Average house price by bedrooms, grade, bathrooms, condition and waterfront

3.1 Average House Price by Bedrooms (Column Chart)



This chart shows the average price for each bedroom count. The price goes up as the bedrooms increase, from about 0.3M for 1 bedroom to a peak of nearly 0.9M at 8 bedrooms. After that it drops, and the house with 11 bedrooms is around 0.5M, so more bedrooms does not always mean a higher price.

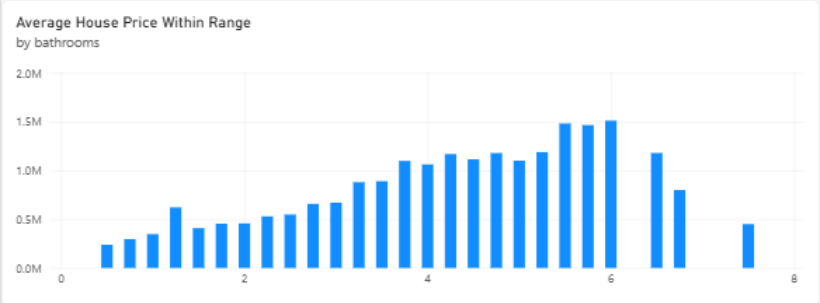
3.2 Average House Price by Grade (Column Chart)



This chart compares the average price with the grade given to the house. The prices stay low (around 0.2M to 0.4M) for the lower grades and then rise quickly after grade 8. The highest

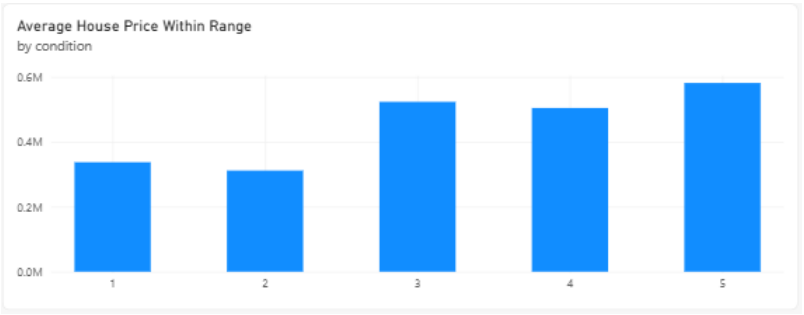
grade, 13, has an average price of about 1.8M. This looks like the strongest relationship among all the charts.

3.3 Average House Price by Bathrooms (Column Chart)



Here the average price is shown for each bathroom count. In general the price increases with the number of bathrooms and reaches about 1.5M around 5.5 to 6 bathrooms. There are some ups and downs in between, and at 7.5 bathrooms the price is low again, probably because there are very few such houses.

3.4 Average House Price by Condition (Column Chart)



This chart shows the average price for condition ratings 1 to 5. Condition 1 and 2 have the lowest prices (about 0.3M), conditions 3 and 4 are around 0.5M, and condition 5 is the highest at about 0.58M. So a better condition gives a higher price, but the difference is not as big as with grade.

3.5 Average House Price by Waterfront (Column Chart)



This chart compares houses that are not on the waterfront (0) with the houses that are (1). Waterfront houses cost about 1.1M on average, which is more than double the price of the others (about 0.5M). This shows that a waterfront view adds a lot of value.

3.6 Price Range (Slider)



This slider is used to set the price range (it is set at 2150000 in the picture). All five charts above are filtered by it, and that is why their titles say "Within Range". We can move it to check how the averages change for cheaper or costlier houses.

4. Conclusion

From this project I learned how to build an interactive dashboard in Power BI with cards, a map, charts and slicers. The cards show that the average house price is 540.65K while the median is 450.00K, so the data is slightly skewed by expensive houses. The map shows that most of the sales are around Seattle and the nearby cities.

The charts show that grade, bathrooms, waterfront and bedrooms have a clear effect on the price, and grade and waterfront have the biggest effect. Condition also increases the price but only a little. Overall the dashboard is easy to use because the sliders let us filter the data and see the results immediately. This helps a buyer or a seller to understand what affects the house price in this area.